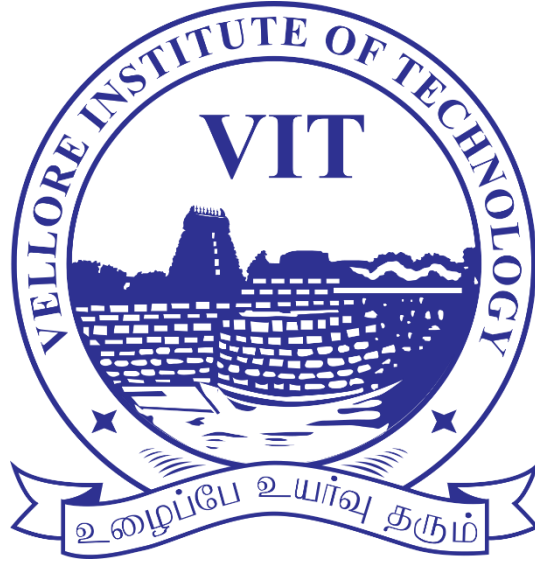




Digital Assignment – 4

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Course Name: MDI3003 – Advanced Predictive Analytics (Lab)

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Probabilistic Customer Segmentation and Segment Prediction Using Demographic, Psychographic, and Behavioral Data with Naive Bayes Classifiers

GitHub Repository:

<https://github.com/raghavmri/23MID0078-advanced-predictive-analytics>

1) Aim & Objectives

The aim of this experiment is to predict predefined customer segments using customer demographic, psychographic and behavioral information. Different Naive Bayes models are implemented and compared to determine which model performs best for the given customer segmentation problem. The main objectives of this experiment are to understand the dataset, examine its quality, perform exploratory data analysis, preprocess the data correctly and build classification models. The models are evaluated using cross validation and a separate test set. The experiment also focuses on understanding which customer segments are difficult to predict, analysing incorrect predictions and studying the confidence of the model.

Another important objective is to ensure that customer segmentation is performed responsibly. Customer predictions should be treated as decision

support and should not be used as the sole basis for sensitive or discriminatory decisions.

2) Dataset Description & Quality Analysis

The dataset used for this experiment contains information about customers along with their predefined customer segment. The task is therefore treated as a supervised classification problem, where the existing customer segment is used as the target variable. The dataset contains different types of customer information. Demographic information can include attributes such as age, education, occupation, income and family size. Psychographic information represents preferences, attitudes and lifestyle related characteristics. Behavioral information represents actions such as purchase frequency, spending, campaign response or discount usage. Before building the models, the dataset was examined to understand its structure and overall quality. The number of records, number of features, target variable and number of customer segments were identified. The dataset was also checked for missing values and duplicate records. Numerical variables were examined for unusual or unrealistic values, while categorical variables were checked for unexpected or inconsistent categories. The distribution of the target classes was also examined to determine whether some customer segments were much larger than others.

Direct identifiers such as customer IDs were not used as predictive features. The dataset was also examined for possible data leakage. In particular, attention was given to whether any feature could directly reveal the target segment or contain information that would not normally be available when making a prediction.

Dataset Quality Findings



Fig 1 Customer Segment Distribution

3) Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the characteristics of the customers and identify patterns that may help distinguish the different customer segments. The numerical variables were first examined to understand their distributions. Histograms and boxplots were used where appropriate to identify the general spread of the variables and detect unusual observations. Categorical variables were then analysed by looking at their frequency distributions. This helped identify which categories were common and whether certain categories appeared more frequently in particular customer segments. The features were also compared across customer segments. This was useful for understanding how the segments differ from each other.

For example, differences in age, spending behaviour, family size, profession or other available customer characteristics can provide an indication of why

some customers are assigned to different segments. Relationships between important numerical variables were also examined where meaningful. A correlation plot or scatter plot can be used when it provides useful information rather than simply adding another figure.

EDA and Segment Findings

Loaded file: business_email_intent.csv
Shape: (220, 4)

	email_id	subject	body	label
0	D1_0001	Please provide	Please provide the latest quarter financial re...	request
1	D1_0002	Can you	Can you send me access permissions for the rep...	request
2	D1_0003	Requesting approval	Requesting approval for budget allocation.	request
3	D1_0004	Please provide	Please provide the latest quarter financial re...	request
4	D1_0005	Can you	Can you send me access permissions for the rep...	request

Fig 2 EDA Analysis O/P

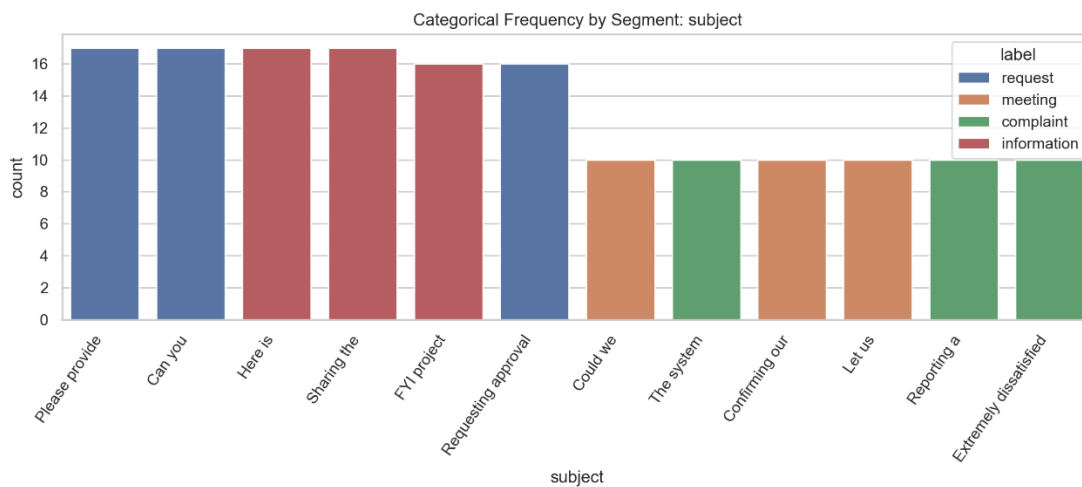


Fig 3 Categorical Frequency

4) Data Preprocessing & Train Test Split

Before training the models, the dataset was preprocessed to make the features suitable for machine learning. Features that were not useful for prediction, such as direct customer identifiers, were removed. Missing values were handled using appropriate imputation techniques. Categorical features were converted into numerical representations suitable for the corresponding Naive Bayes model. Different Naive Bayes algorithms require different types of feature representations. Therefore, model specific preprocessing was applied where necessary. Numerical features were prepared for Gaussian Naive Bayes, while categorical and binary representations were prepared for Categorical Naive Bayes and Bernoulli Naive Bayes respectively. The dataset was divided into training and testing portions using a stratified split so that the distribution of customer segments remained reasonably consistent between the two sets.

An important part of preprocessing was preventing data leakage. Preprocessing transformations were fitted only using the training data or training folds. The test set was kept separate until the final model had been selected. This ensures that the final test performance reflects the model's ability to work with unseen data.

5) Model Implementation

Four classification approaches were implemented as part of the experiment. First, a DummyClassifier was used as a baseline. This model provides a simple reference point and helps determine whether the machine learning models are actually learning useful patterns from the data. Gaussian Naive Bayes was then implemented for numerical feature representations. It assumes that the numerical features follow a Gaussian distribution within each class. Bernoulli Naive Bayes was implemented using binary feature representations. It is useful

when features represent the presence or absence of particular characteristics. Categorical Naive Bayes was implemented for categorical features after converting the categories into suitable numerical representations.

The models were trained using consistent preprocessing and evaluation procedures so that their performance could be compared fairly. The laboratory manual specifically identifies DummyClassifier, GaussianNB, BernoulliNB and CategoricalNB as the core models for this experiment.

6) Model Evaluation & Comparison

The models were evaluated using five fold Stratified Cross Validation. The same folds were used for all models so that the comparison was fair. Accuracy, Macro Precision, Macro Recall, Macro F1 and Weighted F1 were considered during evaluation. Macro F1 was given particular importance because it gives equal importance to each customer segment and is therefore useful when the class distribution is uneven.

The Macro F1 score from each fold was recorded along with the mean and standard deviation. This helped determine not only which model performed well on average, but also how consistent its performance was across different folds. The manual specifically recommends reporting the five fold Macro F1 values, their mean and standard deviation.

Model Comparison

After completing cross validation, all models were compared using a single consolidated comparison.

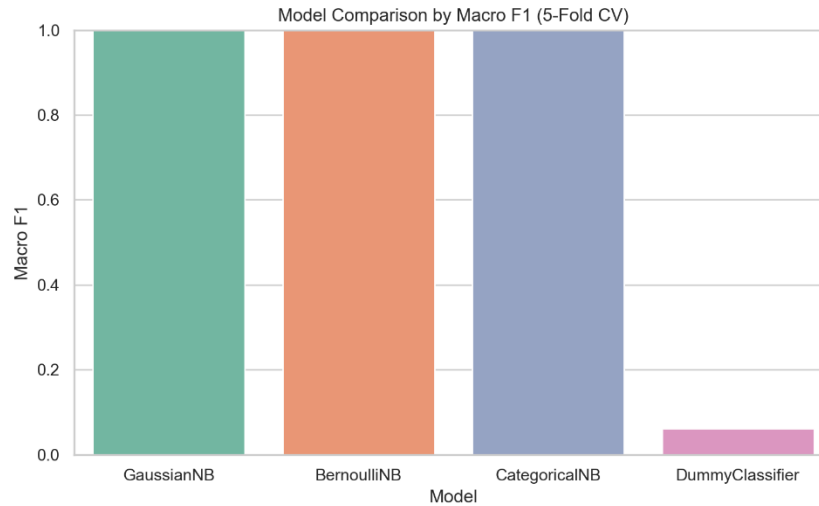


Fig 4 Model Comparison

Model Selection

Based on the cross validation results, **GaussianNB** was selected as the final model. The decision was based mainly on Macro F1, consistency across folds and class wise performance rather than accuracy alone. The final model was selected before evaluating the locked test set.

This directly addresses the requirement that model selection should consider Macro F1, variability, class wise recall and model assumptions rather than simply choosing the highest accuracy.

7) Feature Group Analysis

The contribution of different types of customer information was analysed using a feature group comparison. Four experiments were considered. The first used only demographic features. The second used only psychographic features. The third used behavioral features. The final experiment used all available feature groups together. The same core model and the same cross validation folds

were used for these experiments so that the effect of the feature groups could be compared fairly.

Feature Group Findings

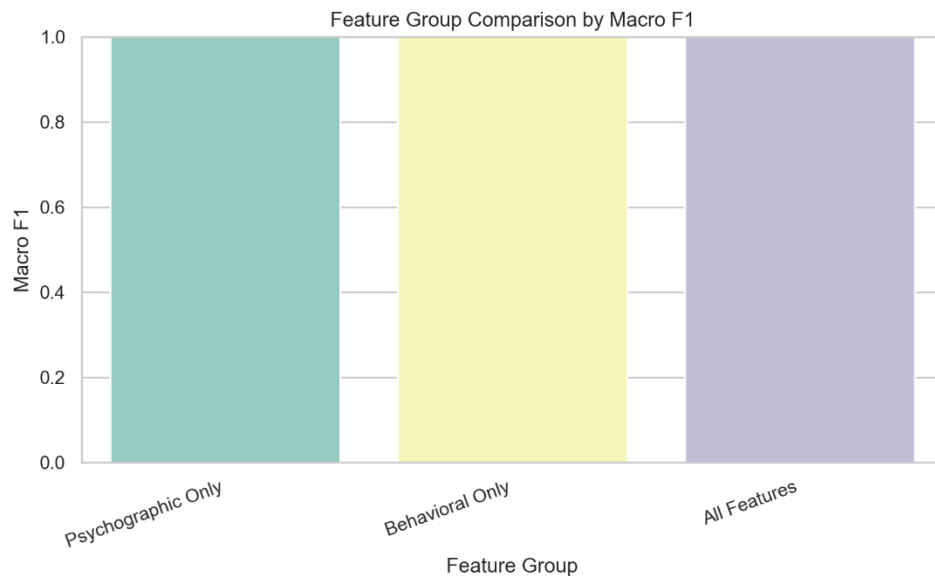


Fig 5 Feature Group Findings

8) Test Evaluation & Diagnostic Analysis

After selecting the final model using cross validation, the model was trained using the complete training dataset and evaluated on the previously locked test set. The final evaluation included Accuracy, Macro Precision, Macro Recall, Macro F1 and Weighted F1. Class wise Precision, Recall and F1 were also examined to understand how well the model performed for each customer segment. The confusion matrix was used to identify which customer segments were correctly classified and which segments were frequently confused with one another.

Diagnostic Analysis

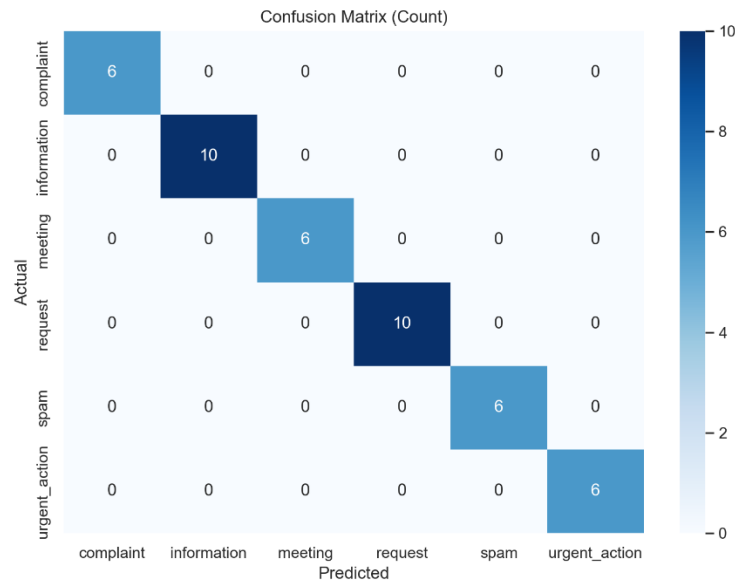


Fig 6 Confusion Matrix

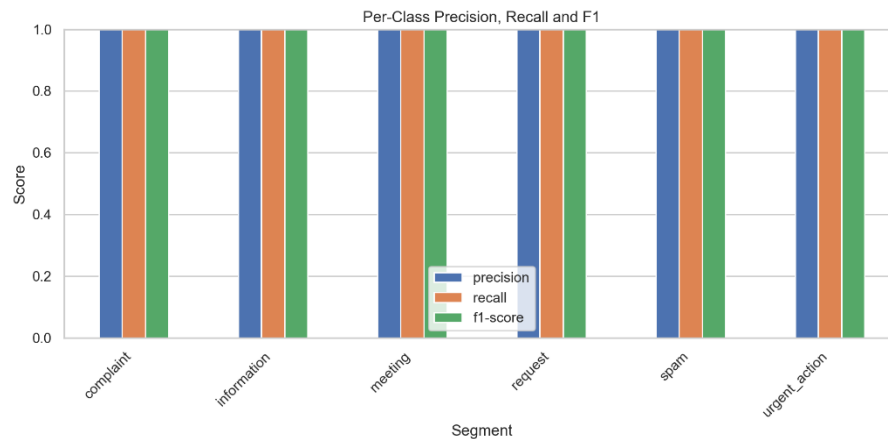


Fig 7 Per Class Stats

9) Confidence & Error Analysis

The probability outputs of the selected model were analysed to understand how confident the model was about its predictions. The highest predicted

probability for each customer was treated as the model's confidence. These probabilities were not interpreted as absolute certainty because model probabilities can sometimes be poorly calibrated.

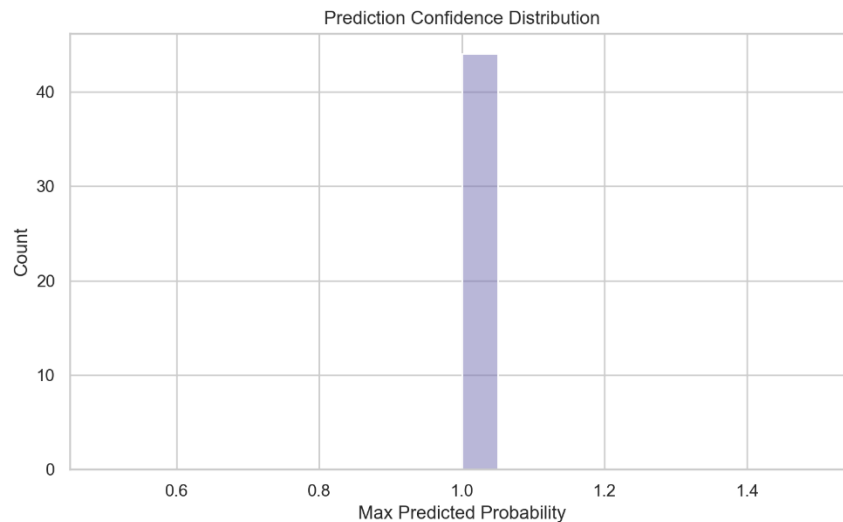


Fig 8 Confidence Distribution

The error analysis then focused on incorrectly classified customers. At least five representative errors were examined to understand why the model made these mistakes. For each selected error, the actual segment, predicted segment, confidence and possible reason for the error were considered. The possible business consequence of the error was also discussed.

10) New Customer Prediction & Responsible Analytics

The final model was used to demonstrate prediction for new customer profiles. Before making a prediction, the input values were checked to ensure that they followed the same structure and valid ranges as the training data. The

preprocessing pipeline was then applied to the new profile before passing it to the selected model.

The prediction output includes the predicted customer segment, probability distribution across the available segments and the model's confidence. A review recommendation can also be provided when the prediction confidence is below the selected threshold.

Responsible Analytics

Customer segmentation can be useful for campaign planning, customer engagement, retention analysis and service personalization. However, the predictions should not be treated as confirmed customer preferences. The model should not be used as the sole basis for discriminatory pricing, denial of services, exclusion or manipulation. Direct identifiers should not be included unnecessarily, and sensitive customer attributes should be handled carefully. Historical segment labels may also contain existing organizational bias. Similarly, psychographic characteristics should not be treated as definitive statements about a person's personality or preferences. Human review should therefore be maintained for decisions that could significantly affect customers. The model should be considered a decision support tool rather than an autonomous decision maker.

11) Results & Discussion

The experiment successfully demonstrated the application of Naive Bayes classification techniques for customer segment prediction. The dataset was examined for its structure, quality, missing values, class distribution and customer characteristics before applying the machine learning models. The exploratory analysis helped identify differences between the customer

segments and provided useful insights into the features associated with each segment. The model comparison showed that the Naive Bayes classifiers were able to learn meaningful patterns from the customer data, although their performance varied across the different segments. The feature group analysis also showed that demographic, psychographic and behavioral features contributed differently to the classification task. The combined feature set provided a broader representation of customer characteristics and helped assess the overall predictive value of the available information. The confusion matrix and error analysis showed that some segments were more difficult to classify than others. Most of the misclassifications occurred between segments with similar characteristics. This highlights the importance of analysing class wise performance rather than relying only on overall accuracy.

The confidence analysis further showed that some predictions were more uncertain than others. Low confidence predictions can therefore be considered for additional review before being used for business decisions. Overall, the results indicate that Naive Bayes provides a useful and relatively simple approach for customer segment classification. However, model performance should be interpreted together with the dataset quality, class distribution, feature characteristics and observed classification errors.

12) Conclusion

The experiment provided a complete workflow for predicting customer segments using supervised machine learning. The process covered dataset quality analysis, exploratory data analysis, preprocessing, implementation of multiple Naive Bayes models, cross validation, model comparison, test evaluation and error analysis. The analysis showed that customer segments can be distinguished to some extent using the available demographic, psychographic and behavioral characteristics. At the same time, overlap between certain segments makes some predictions more challenging.

The experiment also demonstrated the importance of using appropriate evaluation metrics, analysing individual classes and examining prediction errors rather than depending only on overall accuracy. The final model can be useful as a decision support tool for customer segmentation, provided its limitations are understood and human judgement is retained for important business decisions.

13) Appendices

Appendix A: Environment Specifications

The experiment was implemented using Python in a Jupyter Notebook environment. The main libraries used include pandas and NumPy for data handling, scikit learn for preprocessing, model implementation and evaluation, and matplotlib and seaborn for data visualization. The experiments were executed on a standard computing environment without requiring specialized hardware acceleration.

Appendix B: Model Configuration

The Naive Bayes models were implemented using the standard configurations provided by scikit learn. The experiment included DummyClassifier, GaussianNB, BernoulliNB and CategoricalNB. Preprocessing and feature transformations were applied according to the requirements of each model. Five fold Stratified Cross Validation was used for consistent model comparison.

Appendix C: Additional Results

This appendix contains supporting outputs from the experiment that are not included in the main report. These include detailed cross validation results, complete classification reports, additional confusion matrix outputs, feature group comparison results and selected new customer prediction results.

Appendix D: Academic Integrity & Assistance Disclosure

The experiment, analysis and final report were prepared in accordance with the applicable academic integrity requirements. External libraries and datasets used during the experiment have been appropriately acknowledged. Any AI assistance used during the preparation of the report or debugging of code was treated as a supporting tool, while the experimental results, interpretations and final submission were reviewed and verified by the authors